

Advanced Calculus

Differential Equations (DE):

An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a differential equation.

Ordinary Differential Equations (ODE):

A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an ordinary differential equation.

Partial Differential Equation (PDE):

A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variable is called a partial differential equation.

EXAMPLES BY CATEGORY

Ordinary Differential Equations (ODE)

$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$$

$$\frac{d^4x}{dt^4} + 5\frac{d^2x}{dt^2} + 3x = \sin(t)$$

Partial Differential Equations (PDE)

$$\frac{\partial v}{\partial \sigma} + \frac{\partial v}{\partial t} = v$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Not DE but Basic Derivatives

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Order of a DE:

The order of the highest-order derivative involved in a differential equation is called the order of the differential equation.

Degree of a DE:

The degree of a differential equation is the power of the highest-order derivative present in the equation.

EXAMPLES BY CATEGORY

1st Order 1st Degree ODE

$$\frac{dy}{dx} + y = e^x$$

2nd Order 1st Degree ODE

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

2nd Order 2nd Degree ODE

$$\left(\frac{d^2y}{dx^2}\right)^2 + y\frac{dy}{dx} = 0$$

1st Order 1st Degree PDE

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0$$

2nd Order 1st Degree PDE

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

2nd Order 2nd Degree PDE

$$\left(\frac{\partial^2 u}{\partial x^2}\right)^2 + \left(\frac{\partial^2 u}{\partial y^2}\right)^2 = 0$$

Linear ODE:

A linear ordinary differential equation of order n , in the dependent variable y and the independent variable x , is an equation that can be expressed in the form

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_n y = f(x)$$

or equivalently,

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n(x) y = f(x)$$

where $a_0, a_1, \dots, a_n$ are functions of x (not identically zero) and $f(x)$ is a given function.

Important Observations:

- 1) The dependent variable y and its various derivatives occur to the first degree only.
- 2) No products of y and/or any of its derivatives are present.
- 3) No transcendental functions of y or its derivatives occur (e.g., $\sin y$, e^y).

Nonlinear ODE:

A nonlinear ordinary differential equation is an ordinary differential equation that is not linear.

EXAMPLES BY CATEGORY

Linear ODE with Constant Coefficients

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

Linear ODE with Variable Coefficients

$$\frac{d^4y}{dx^4} + x^{30}\frac{dy}{dx} + 2y = xe^x$$

Nonlinear ODE Examples

$$\left(\frac{d^2y}{dx^2}\right)^2 + y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \sin(y) = 0$$

$$A\frac{d^2y}{dx^2} + K\frac{d^3y}{dx^3} + x\frac{dy}{dx} = xe^x\frac{dx}{dy}$$

General Form of a Nonlinear ODE

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0$$

What Makes an ODE Nonlinear?

An ODE is classified as nonlinear if it violates any of the key linear criteria:

- **Higher Powers:** Any y or derivative term has an exponent other than 1.
- **Product Terms:** The variable y is multiplied by its own derivatives.
- **Nonlinear Functions:** The equation contains functions like $\sin(y)$, e^y , or $\ln(y)$.

Origin and Application:

Differential equations occur in connection with numerous problems encountered across various branches of science and engineering:

- Population Dynamics
- Dynamical Systems and Chaos
- Fluid Dynamics
- Heterogeneous Heat Transfer
- Biophysics and Mathematical Biology

Formation of Differential Equations:

To form a differential equation from a given relation involving independent/dependent variables and arbitrary constants, we eliminate the arbitrary constants by successive differentiation.

Example Problem:

Form the differential equation by eliminating the constants a and b from:

$$y = ax + bx^2 \quad (1)$$

Solution:

Given the relation with two arbitrary constants a and b :

$$y = ax + bx^2 \quad (2)$$

Since there are two arbitrary constants, we differentiate the equation twice with respect to x to eliminate them.

Step 1: First Differentiation Differentiating Equation (2) with respect to x :

$$\frac{dy}{dx} = a + 2bx \quad (3)$$

Step 2: Second Differentiation Differentiating Equation (3) with respect to x :

$$\frac{d^2y}{dx^2} = 2b \implies b = \frac{1}{2} \frac{d^2y}{dx^2} \quad (4)$$

Step 3: Eliminate Constant a Substitute the value of b from Equation (4) into Equation (3):

$$\begin{aligned} \frac{dy}{dx} &= a + 2 \left(\frac{1}{2} \frac{d^2y}{dx^2} \right) x \\ \frac{dy}{dx} &= a + x \frac{d^2y}{dx^2} \implies a = \frac{dy}{dx} - x \frac{d^2y}{dx^2} \end{aligned}$$

Step 4: Substitute a and b into the Original Equation Substitute the expressions for a and b back into Equation (2):

$$y = \left(\frac{dy}{dx} - x \frac{d^2y}{dx^2} \right) x + \left(\frac{1}{2} \frac{d^2y}{dx^2} \right) x^2$$

Expand the terms:

$$y = x \frac{dy}{dx} - x^2 \frac{d^2y}{dx^2} + \frac{1}{2} x^2 \frac{d^2y}{dx^2}$$

Combine the like terms of $\frac{d^2y}{dx^2}$:

$$y = x \frac{dy}{dx} - \frac{1}{2} x^2 \frac{d^2y}{dx^2}$$

Final Form: Multiply the entire equation by 2 and rearrange terms to obtain the standard form:

$$x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0 \quad (5)$$

This is the required ordinary differential equation of 2^{nd} order and 1^{st} degree.